

- ① prove that the open balls are open sets
- ② write the generating function for bernoulli polynomials
- ③ check for convergence

- ① $\lim x_n = \frac{1}{p_n(m^2)} + \frac{1}{p_n(m^3)} + \dots + \frac{1}{p_n(m^m)}$

- ② Wallis Formula (in dem)

- ③ check for convergence $\sum \left(\frac{n!}{((a+1)(a+2)(a+n))} \right), a > -1$

- ① limita de π

- ② seria maclaurin pt tangenta

- ③ seria maclaurin pt $\frac{x}{x^4 + x^2 + 1}$

- ① def: topologie, multime deschisă, multime închisă, interior

- ② $\lim_{n \rightarrow \infty} \left(\frac{1}{1+n} + \frac{1}{2+n} + \dots + \frac{1}{2n} \right)$

- ③ maclaurin pt $\ln(1 + n + n^2)$



- ① Def: metrice, spatii metrice, bile într-un spațiu metric

- ② convergența și raza seriei $n! \cdot x^n$

- ③ maclaurin pt $\arctg x^2$

- ① Def: vecinătate, aderență, derivata unei multimi

- ② natura seriei $\frac{(a+1)(a+2)\dots(a+n)}{n!}$

- ③ maclaurin pt $\log x + \frac{1}{x-1}$

- ① Banach's fixed point theorem + dem

- ② Ext into Maclaurin series of $\sqrt{(1+x)(1-x)}$ and find

the relation with $\arctan x$

- ③ convergence for $a^m \left(1 + \frac{1}{m} \right) (a m^2 + b m)$